

			
	Ma'aden Phosphate 3 Phase I		
Project Phase :	Stage 4 - EPCM	Ma'aden Project N° :	18-PME-PH3-PH1-001
Main Title:	FENCING AND GATES SPECIFICATION	Document N°:	MD-520-1000-EG-CV-SPC-0006
		Revision:	C01

ISSUED FOR CONSTRUCTION

MA'ADEN

Phosphate 3 Phase I – Stage 4

Fencing and Gates Specification

		Ma'aden Phosphate 3 – Phase 1	
NAME	TITLE	SIGN	DATE
REVIEW GRADE			
Result	Code	Description	
☐	A	Accepted	
☐	C	Proceed with exceptions to next Submission and Revision	
☐	R	Do Not Proceed - Revise as noted and Resubmit	
Acceptance in any of these categories in no way relieves the Contractor/Supplier of its responsibility for the due and proper performance of the Works in accordance with the Contract/Purchase Order.			
PROJECT MANAGER SIGN-OFF:			
NAME			DATE

Rev.	Issue Date	Revision Description	Prepared by	Checked by	Discipline Approved	Project Approved by
A01	08 Nov 23	ISSUED FOR REVIEW	VSS	SHB	SHB	AMN
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1. INTRODUCTION

1.1 Project Introduction

The Saudi Arabian Mining Company (Ma'aden) is proceeding with the development of two strategic projects:

1-Ras Al Khair project: Two Diammonium phosphate (DAP) plants (phase 1) with all associated facilities / utilities and storage in the area adjacent to the Eastern side of Ma'aden Phosphate Company (MPC) at Ras Al-Khair

2- Wa'ad Al Shamal project: the proposed site includes a mine and related infrastructure works as well as chemical plants associated with the processing of phosphate in the production of fertilizers including related infrastructures such as roads and railways etc. located within 5km from the mine area. The project comprises also a dam for flood protection and a gypsum storage.

1.2 Project Description

1.2.1 Ras Al Khair project (RAK)

The scope of service for RAK describes without limitation the design, procurement and building of an overall chemical complex for the production of Fertilizers.

The scope of work covers the following plants and facilities

- Granulation Plant: 2 Trains of 1 million MTPY (DAP Equivalent) per train to produce different fertilizers grades.
- Material Handling and Storage Halls.
- Connections at the battery limits
- All other utilities, equipment and material required to ensure safety, operability, maintainability, process performances and reliability for all the plants life cycle.

1.2.2 Wa'ad Al Shamal project (WAS)

The scope of service for WAS concerns the design, procurement and building of an overall chemical complex for the whole Phosphoric acid production value chain, starting from the mining crushing area to the delivery of MGA at the battery limits.

The scope covers the following plants and facilities

- Crushing area, including its electrical substation
- Beneficiation Plant of 2.05 Million MT / year, Technology provided by FLS
- Phosphate tailing facility
- SAP of 1.75 Million MT / year, licensed by Metso: Outotec
- Power Plant of approximately 65 MW

			
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- PAP of 0.5 Million MTP2O5 / year, licensed by Prayon
- Gypsum Storage facility
- Water Treatment area
- Connections at the battery limits
- All other utilities, equipment and material required to ensure safety, operability, maintainability, process performances and reliability for all the plants life cycle.

1.3 Scope

This document identifies the basic requirements and provides the scope of work for the security fencing systems to be implemented for Ma'aden Phosphate-3 Facility at Ras Al Khair (RAK) Waad Al Shamal (WAS) areas.

The security fencing system shall be in full compliance with applicable HCIS SEC directives, Ma'aden standards, local laws and codes, in addition to the applicable international standards.

This document also defines the relevant codes, standards, and references to be used in the design and construction of security fencing system for the detail design phase. It should be understood here that this document does not include all the required and detailed information to complete a detail design of the systems. However, the document, as per project objectives, shall define the basis for specifying and defining "what" Security fencing System and the requirements needed to achieve a fully operational plant and satisfy project owners.

Due to the nature of the project, quality and reliability of the security fencing system to be provided are very important. The systems shall therefore be of updated technology, proven and reliable design, complying with the applicable system specifications, recommendations, and standards.

This document shall be used as an overall guidance for the design, construction and installation of the Security fencing system and services for the Ras Al Khair (RAK) and Wa'ad Al Shamal (WAS) facilities under Phosphate 3 Phase I Project.

1.4 Purpose

The purpose of this specification is to set minimum engineering standard for security fencing system for all Ma'aden projects.

			
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2. DEFINITION AND ABBREVIATIONS

2.1 Definitions

Table 2.1- Definition

NAME	DESCRIPTION
CONTRACTOR	An entity, engaged or approved by the employer, entrusted with a contract to perform work or cause work to be performed.
Work	The entire work and services and all things which the contractor is or may be required to execute, supply and do under the contract having regard to the scope of project, or to cause to be executed, supplied or done, for the purpose of constructing and commissioning the plant
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
EPCM Contractor	Worley
Manufacturer or Vendor	VENDOR responsible for design, sourcing of materials, manufacturing, assembling, testing and supply of equipment as per this specification and associated drawings and datasheets. The wording Manufacturer is a synonymous of the term VENDOR or Supplier.
Site	Geographical location where Services are to be performed.

			
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2.2 Abbreviations

Table 2.2- Abbreviations

Acronym	Description
AVB	Anti-Vehicle Barrier
DD	Detail Design
FSC	Facility Security Classification
HCIS	High Commission for Industrial Security
NCA	National Cybersecurity Authority
RAK	Ras Al-Khair
SEC	HCIS Security Directives
SRA	Security Risk Assessment
WAS	Wa'ad Al Shamal
ASTM	American Society for Testing and Materials
SBC	Saudi Building Code
RWM	Rigid Welded Mesh

			
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3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the revision of the relevant standards or codes mentioned in structural design criteria of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

The work shall conform to the following applicable codes and standards. Any conflicting requirements among the specifications, codes and standards referred; the most stringent requirement will be adopted after obtaining formal approval from Ma'aden.

3.2.1 Local Codes and Standards

Ministry of Interior

- HCIS Security Directives for Industrial Facilities
- HCIS Directive SEC-02 2017, Security Fencing System

3.2.2 International Codes and Standards

American Society for Testing and Materials (ASTM)

ASTM A121	Standard Specification for Metallic-Coated Carbon Steel Barbed Wire
ASTM A123	Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A123M	Zinc Coatings on Iron and Steel Products
ASTM A529	Standard Specification for Structural Steel With 42 000 Psi (290 Mpa) Minimum Yield Point, and 12.7 mm Maximum Thickness.
ASTM A615	Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement.

			
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ASTM A641	Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire
ASTM A392	Standard Specification for Zinc-Coated Steel Chain-Link Fence Fabric
ASTM A853	Standard Specification for Steel Wire, Carbon, for General Use
ASTM C595	blended hydraulic cements for both general and special applications, using slag or pozzolan, or both, with 8ortland cement or 8ortland cement clinker or slag with lime
ASTM F567	Standard Practice for Installation of Chain-Link Fence
ASTM F552	Standard Terminology Relating to Chain Link Fencing
ASTM F1083	Standard Specification for Pipe, Steel, Hot-Dipped Zinc-Coated (Galvanized) Welded, for Fence Structures
ASTM F626	Standard Specification for Fence Fittings
ASTM F668	Standard Specification for Polyvinyl Chloride (PVC) and Other Organic Polymer-Coated Steel Chain-Link Fence Fabric
ASTM F934	Standard Specification for Standard Colors for Polymer-Coated Chain Link Fence Materials
ASTM F1043	Standard Specification for Strength and Protective Coatings on Steel Industrial Chain Link Fence Framework
ASTM F1664	Standard Specification for Poly Vinyl Chloride (PVC) and Other Conforming Organic Polymer-Coated Steel Tension Wire Used with Chain-Link Fence
ASTM F1665	Standard Specification for Poly(Vinyl Chloride) (PVC) and Other Conforming Organic Polymer-Coated Steel Barbed Wire Used With Chain-Link Fence
ASTM F1910	Standard Specification for Long Barbed Tape Obstacles
ASTM F1911	Standard Practice for Installation of Barbed Tape
ASTM F2453	Standard Specification for Welded Wire Mesh Fence Fabric
ASTM F2781 - 10	Standard Practice for Testing Forced Entry, Ballistic and Low Impact Resistance

			
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3.2.3 Ma'aden Standards

MD-101-SMEM-EG-CV-CRT-0001	Civil Design Criteria - General
MD-101-SMEM-EG-CV-SPC-0001	Earthworks
MD-101-SMEM-EG-CV-SPC-0009	Geotechnical Investigation
MD-101-SMEM-EG-ST-SPC-0005	Concrete Works
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

3.2.4 Project Standards

MD-520-1000-EG-CV-SPC-0001	Earthwork Specification
MD-520-1000-EG-CV-SPC-0002	Concrete Works specification.
MD-520-1000-EG-CV-CRT-0001	Civil and Structural Design Criteria
MD-521-1000-EG-CV-CRT-0001	Design Criteria INFRA - WAS
MD-522-1000-EG-CV-CRT-0001	Design Criteria INFRA – RAK

4. FENCING AND GATE SPECIFICATIONS

4.1 Category 1 and 2 Fencing System

- Category 1 & 2 Fencing systems will be provided for this facility as per HCIS Directive SEC-02.
- Category-1 shall be provided for RAK Area A and shall be integrated with Ammonia -03 existing Category-1.
- Category-2 fence shall be provided for WAS Block B Area.
- Category 1 & 2 fencing system will consist of following:

4.1.1 Anti-Personnel Fence Area

- The anti-personnel fence shall consist of a 3m high fence configured with three rolls of concertina wire.
- A roll of concertina wire shall be mounted on top of the fence with a second roll mounted on the front top and the third roll mounted on the external bottom of the fence
- The anti-personnel fence for Category 1 fence shall be with any Anti vehicle barrier (AVB).
- Category-1 & 2 fence will be of Rigid Weld Mesh Panel and will comply HCIS Directive SEC-02.

			
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- All fence materials including fabric, post and fittings coatings will conform to HCIS Directive SEC-02.
- The anti-personnel fence will be tested, validated, and certified to provide forced entry resistance as specified in HCIS Directive Sec-02 Section 6.4.
- All fence and gate posts will have minimum yield strength of 42000 psi (290 mpa) and will meet ASTM A529 requirements as a minimum.
- The fencing panels will be attached to the line posts at intervals not exceeding 200mm along each post with locking plates, bolts, nuts and washers.

4.1.2 Anti-Vehicle Barrier (AVB)

- The AVB shall be installed adjacent to the inside of the Anti-Personnel Fence.
- It shall be crash rated, certified and installed as specified in HCIS Directive SEC-06.
- The AVB shall be non-obscuring to allow security patrols and perimeter monitoring and assessment systems to have a clear field of view through the AVB and anti-personnel fence.

4.1.3 Fence Material specification

4.1.3.1 Durability Performance

- Fence panels, and associated components, will be galvanized in accordance with ASTM A641 requirements then fusion bond/polyester coated in accordance to ASTM F668. Galvanizing will be applied after welding.
- Terminal posts, corner posts, line posts, support braces, brace post, top rail, truss rods to be internally and externally galvanized in accordance to ASTM F1043 and then fusion bond/polyester coated internally and externally in accordance to ASTM F668.
- A minimum coating thickness of 350 micron will be applied.
- The color of the PVC will be dark green or black as per ASTM F934 & HCIS SEC-02 Security Fencing Systems
- Coatings damaged in the field during installation or operations will be repaired using methods and techniques recommended by the manufacturer.

4.1.3.2 Fence Fabric and Panels

- Anti-personnel fences will be constructed of Rigid Weld Mesh panel.
- The fence will have an aperture of 75mm x 12.5mm $\pm 2\%$ center to center. Vertical reinforcing bars, 15mm wide x 5mm thick $\pm 2\%$, will be placed at 75mm $\pm 2\%$ intervals. The fence panel will be constructed of vinyl-coated machine welded mesh using 3mm $\pm 2\%$ diameter wire unless otherwise specified.
- Rigid Weld Mesh panel widths will be between 3m & 3.5m.

			
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4.1.3.3 Visibility Index (VI)

- The Visibility Index (VI) may not exceed 45%. This requires that no more than 45% of the fence panel area may be covered with fence wiring and strips.

4.1.3.4 Razor Tape

Razor Tape (i.e. concertina) materials and configurations shall be in accordance with ASTM F1910 with the following specific provisions:

- Razor tape shall typically be 600 mm diameter (+/- 50 mm). Each loop shall consist of 24(+/- 1) clusters of four needle-sharp barbs on 100mm centers, each barb measuring a minimum of 30mm in length.
- The razor tape shall be secured along its entire length by a minimum 2mm thick rail with spikes on it. Where required, razor tape shall be attached to the fence panel with a minimum 2mm thick bracket at 500mm intervals.
- Where razor tape is placed on the ground, each coil shall be securely anchored to the ground at 1.5 meter intervals.
- The fence posts shall be extended as needed to secure the razor tape.
- Installation of razor tape will follow the practice of ASTM F1911.

4.1.3.5 Barbed Wire

- Six strands of barbed wire shall be installed to secure the razor tape installed on top of the fence.
- Strands shall be spaced uniformly and attached to frame with bands, clips, or eyebolts.
- The strands of barbed wire shall be stretched to remove sag and be anchored firmly to extension arms.

4.1.3.6 Tie Wire

- Tie Wire will have a core of 1.4 mm, minimum, and shall be PVC coated with an outside diameter of 2 mm, minimum.
- Tie wire will comply with the requirements for high security round wire ties specified in ASTM F626.
- Any of the high security tie wire methods specified in ASTM F626 are acceptable.

4.1.3.7 Concrete for Fence

Concrete used for fencing system installation will comply with MD-520-1000-EG-CV-SPC-002 Specification and the following requirements:

- Minimum compressive strength of concrete at 28 days will be 28 Mpa for structural concrete (Maximum water cement ratio will be 0.4) and 14 Mpa for lean concrete.

			
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- Type of cement shall be type-V Sulfate resistant (or) as per geotechnical recommendations.
- Concrete surface protection coatings are as per geotechnical recommendations.
- Before placing components such as gates or RWM panels, the concrete will have cured a minimum of 7 days unless the concrete reaches at least 75 percent of its design strength.

4.1.3.8 Reinforcement for Fence

- All reinforcement steel bars shall be hot rolled grade 60 (ASTM A615) high tensile deformed billet steel bars.
- Fusion bonded epoxy coated rebars will be used for all structural concrete.

4.1.3.9 Electrical Grounding

Grounding will comply with HCIS SEC-02 Section 6.11 and the following specific requirement:

- Fences which are within 10 meter of an enclosed ground grid or ground loop that is connected to equipment operated at 1000 Volt or greater will not be PVC coated and will be grounded at intervals not exceeding 15 meters to the ground grid or loop. All fences within 3 meters of a ground grid or ground electrode will be bonded at the nearest fence post to the ground grid or ground electrode.
- Fences that pass under a transmission line operating at 69 kV and above will not be PVC coated and be grounded at intervals not exceeding 15 meters on that portion of the fence within 100 meter of the power line.
- Fences that cross over a ground grid or conductors that connect two ground grids will not be PVC coated and will have a bond between the grid or conductors and the nearest post. If the crossing area is extensive, the bond is required every 50 meters.

4.1.3.10 Alignment and Terrain Profiling

- Fence will run in straight lines as far as possible. Where changes of direction are required, the angle will not be less than 90 degrees.
- Fences will match the terrain profile. When required due to terrain the fence panel width may be reduced from the nominal 3 meters width requirement and shorter panel widths be used with fence posts increased in length as needed. This may create a stepped profile on top of the fence. The razor tape will seamlessly cover this profile.
- Where the fence crosses feature such as streams and drainage ditches and where conforming of the fence to the ground contour is impractical, the fence will span the depression.

The space below the bottom of the fence will be closed with extra fence panel or barbed wire. If extra length fence posts are required at such locations, they will be furnished and installed in lieu of standard-length posts as may be required.

			
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- A reasonably smooth profile at the fence line will be provided. The bottom of the fence will not be more than 25 mm (1 inch) above the finished ground line.

4.1.3.11 Post Installation Requirements

- All line post installation intervals will match RWM panel size up to a maximum spacing of 3.5 meter.
- Posts will typically be 3.6 meter in length or as required by fencing system design.
- Fence line, corner, end, gate, brace or pull posts will be spaced as needed to guarantee fence stability or as required due to changes in vertical or horizontal alignment.
- Posts to be coated both externally and internally.
- The top of the installed post will be covered with a cap that is secured using either a pop rivet, screw or bolt.

4.1.3.12 Rigid Welded Mesh Specific Requirements

Materials will be uniform and consistent, conform to ASTM F2453 and shall meet the following requirements:

The fence shall be installed in accordance with ASTM F567 and the specific provisions in this section.

- Panels will be secured using bolts (min 5mm thick), nuts and washers for fixing to be anti-tamper and made from Grade 316 or 304 Stainless Steel or hot dip galvanized.
- The fencing panels will be attached to the line posts at intervals not exceeding 200mm along each post with locking plates, bolts, nuts and washers.
- Full locking system will be positioned away from attack on the internal side of fence line.
- Bolts and nuts will be tampered proof.
- The welded mesh fence panel will be installed so that the posts are enclosed.
- If any damage occurs to the fence post or panels during installation, corrosion protection touch up should be applied within 48 hours.
- The bottom of the welded mesh panel will be tied to a continuous concrete anchor and fastened with wire ties at intervals not exceeding 250mm.
- Images are provided for conceptual purpose.

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HCIS Category - 1 & 2 Rigid Welded Mesh Structural details

			
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4.2 Category 3 Fence System (Applicable for RAK Admin and Unloading Area)

CONTRACTOR shall install Category 3 Fencing system as per HCIS 2017 Directive.

Category - 3 fencing system will consist of following.

4.2.1 Anti-Personnel Fence Area

- The anti-personnel fence shall consist of a 3m high fence configured with two rolls of concertina wire
- A roll of concertina wire shall be mounted on top of the fence with the second roll mounted on the front top.
- All fence materials including fabric, post and fittings coatings will conform to HCIS SEC-02 specification and Saudi Aramco Standard.
- Straining posts (corner, end, gate or tension) will be installed at the beginning and end and at a maximum of 90 meters intervals of fence fabric and at changes vertical and horizontal alignments exceeding 22.5° deflection angle. Straining posts will be braced in the plane of fencing to the adjacent line posts.
- All fence and gate posts will have a minimum yield strength of 42000 psi (290 mpa) and will meet ASTM A529 requirements as a minimum.

4.2.2 Fence Material Specification

4.2.2.1 Durability Performance

- Fence panels, and associated components, will be galvanized in accordance with ASTM A641 requirements then fusion bond/polyester coated in accordance to ASTM F668. Galvanizing will be applied after welding.
- Terminal posts, corner posts, line posts, support braces, brace post, top rail, truss rods to be internally and externally galvanized in accordance to ASTM F1043 and then fusion bond/polyester coated internally and externally in accordance to ASTM F668.
- A minimum coating thickness of 350 micron will be applied.
- The color of the PVC will be dark green or black as per ASTM F934. HCIS SEC-02 Security Fencing requirements.
- Coatings damaged in the field during installation or operations will be repaired using methods and techniques recommended by the manufacturer.

4.2.2.2 Fence Fabric

- Anti-personnel fences will be constructed of Chain-Link (CL) fabric.
- Fabric will be vinyl-coated chain link with a typical 50 mm diamond mesh. The standard length of roll will be 15 meters, or greater, and will be woven continuously without splices.

			
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4.2.2.3 Visibility Index (VI)

The Visibility Index (VI) may not exceed 45%. This requires that no more than 45% of the fence panel area may be covered with fence wiring and strips.

4.2.2.4 Razor Tape

Razor Tape (i.e. concertina) materials and configurations shall be in accordance with ASTM F1910 with the following specific provisions:

- Razor tape shall typically be 600 mm diameter (+/- 50 mm). Each loop shall consist of 24(+/- 1) clusters of four needle-sharp barbs on 100mm centers, each barb measuring a minimum of 30mm in length.
- The razor tape shall be secured along its entire length by a minimum 2mm thick rail with spikes on it. Where required, razor tape shall be attached to the fence panel with a minimum 2mm thick bracket at 500mm intervals.
- Where razor tape is placed on the ground, each coil shall be securely anchored to the ground at 1.5 meter intervals.
- The fence posts shall be extended as needed to secure the razor tape.
- Installation of razor tape will follow the practice of ASTM F1911.

4.2.2.5 Barbed Wire

- Six strands of barbed wire shall be installed to secure the razor tape installed on top of the fence.
- Strands shall be spaced uniformly and attached to frame with bands, clips, or eyebolts.
- The strands of barbed wire shall be stretched to remove sag and be anchored firmly to extension arms.

4.2.2.6 Tie Wire

- Tie Wire will have a core of 1.4 mm, minimum, and shall be PVC coated with an outside diameter of 2 mm, minimum.
- Tie wire will comply with the requirements for high security round wire ties specified in ASTM F626.
- Any of the high-security tie wire methods specified in ASTM F626 are acceptable.

4.2.2.7 Top Rail and Tension Wire

- Fence will be installed with a top rail and bottom tension wire.
- The top rail will be continuous except over drive gates. One expansion/contraction coupling will be used for every 30 meters of rail.

If it is necessary for a top rail to be omitted, a top tension wire will be installed.

			
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- The tension wire will be secured to the chain link fabric with high-security round tie wire.
- Tension wire will be secured to terminal posts by means of a winding bracket.
- PVC coated galvanized tension wires shall conform to ASTM F1664, Class 2A.

4.2.2.8 Concrete for Fence

Concrete used for fencing system installation will comply with MD-520-1000-EG-CV-SPC-002 Specification and the following requirements:

- Minimum compressive strength of concrete at 28 days will be 28 Mpa for structural concrete (Maximum water cement ratio will be 0.4) and 14 Mpa for lean concrete.
- Type of cement shall be type-V Sulfate resistant (or) as per geotechnical recommendations.
- Before placing components such as chain link fabric, rails, tension wire and gates, the concrete will have cured a minimum of 7 days unless the concrete reaches at least 75 percent of its design strength.
- Concrete surface protection coatings are as per geotechnical recommendations.

4.2.2.9 Electrical Grounding

Grounding will comply with HCIS SEC-07 and the following specific requirement:

- Fences which are within 10 meter of an enclosed ground grid or ground loop that is connected to equipment operated at 1000 Volt or greater will not be PVC coated and will be grounded at intervals not exceeding 15 meter to the ground grid or loop. All fences within 3 meters of a ground grid or ground electrode will be bonded at the nearest fence post to the ground grid or ground electrode.
- Fences that pass under a transmission line operating at 69 kV and above will not be PVC coated and be grounded at intervals not exceeding 15 meter on that portion of the fence within 100 meter of the power line.
- Fences that cross over a ground grid or conductors that connect two ground grids will not be PVC coated and will have a bond between the grid or conductors and the nearest post. If the crossing area is extensive, the bond is required every 50 meter.

4.2.2.10 Alignment and Terrain Profiling

- Fence will run in straight lines as far as possible. Where changes of direction are required the angle will not be less than 90 degrees.
- Fences will match the terrain profile. When required due to terrain the fence panel width may be reduced from the nominal 3 meters width requirement and shorter panel widths be used with fence posts increased in length as needed. This may create a stepped profile on top of the fence. The razor tape will seamlessly cover this profile.

			
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- Where the fence crosses feature such as streams and drainage ditches and where conforming of the fence to the ground contour is impractical, the fence will span the depression.

The space below the bottom of the fence will be closed with extra fence panel or barbed wire. If extra length fence posts are required at such locations, they will be furnished and installed in lieu of standard-length posts as may be required.

- A reasonably smooth profile at the fence line will be provided. The bottom of the fence will not be more than 25 mm (1 inch) above the finished ground line.

4.2.2.11 Post Installation Requirements

- All line posts will be equally spaced at 3 meters intervals or less.
- Posts will typically be 3.6 meter in length or as required by fencing system design.
- Fence line, corner, end, gate, brace or pull posts will be spaced as needed to guarantee fence stability or as required due to changes in vertical or horizontal alignment.
- Posts to be coated both externally and internally.
- The top of the installed post will be covered with a cap that is secured using either a pop rivet, screw or bolt.

4.2.2.12 Chain Link Specific Requirements

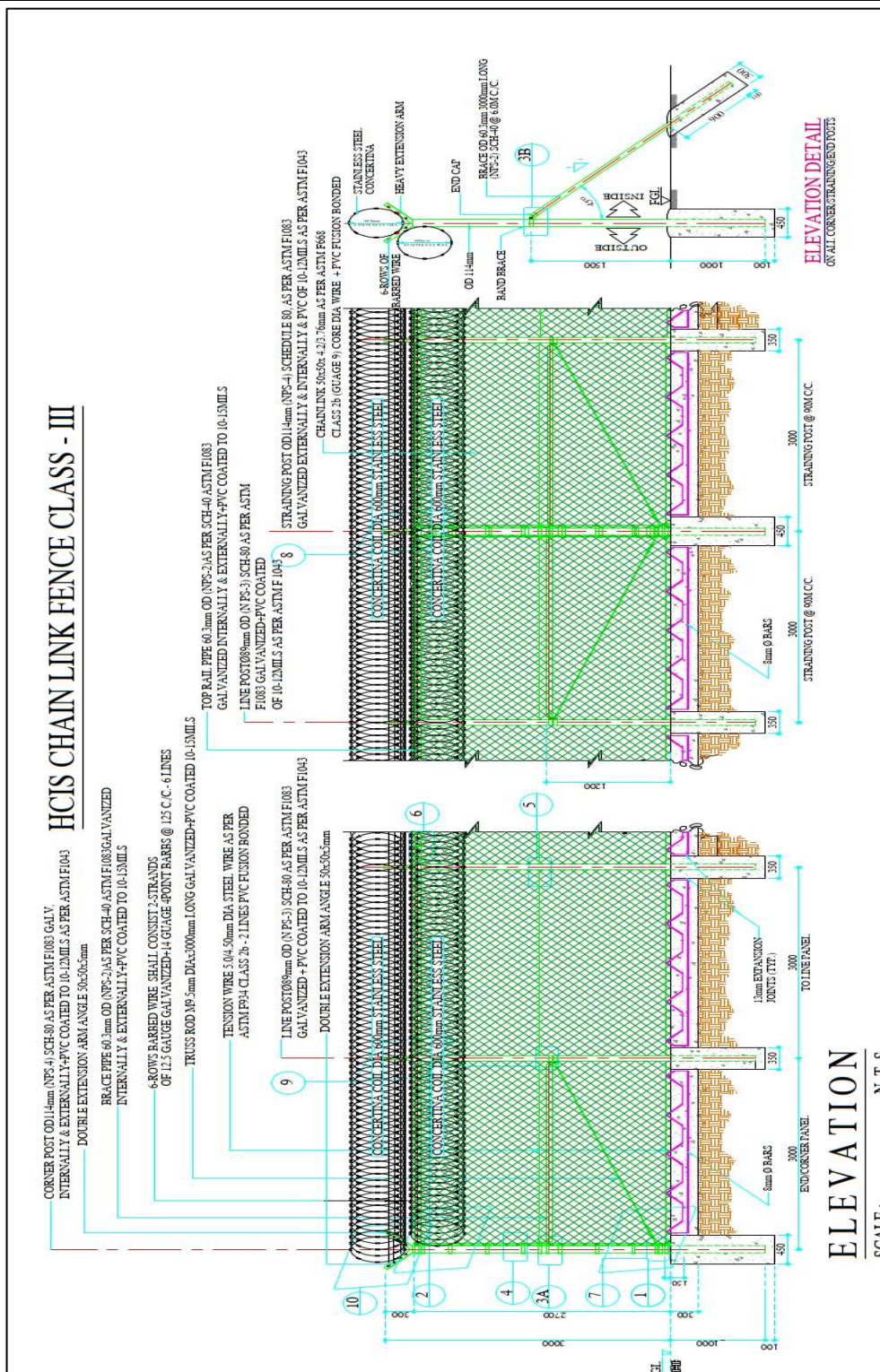
The fence will be installed in accordance with ASTM F567 and the specific provisions in this section.

- The fencing fabric will be attached to all line posts by means of tie wire.
- The chain link fence fabric will be installed so that the posts are enclosed.
- The bottom of the chain link fabric will be tied to a continuous concrete anchor.
- Tension bars will be used with fabric bands.
- The bottom edge of fabric will be fastened to the tension wire with wire ties.
- The fabric will be stretched tautly so as not to deflect more than 76mm (3inch) in the center of the fence panel in between the two-line posts. Fabric will be fastened securely to the posts.

4.2.2.13 Bracing in Plane of Fence:

- Gate and end posts will be braced to the nearest line post with one complete brace assembly.
- Corner and pull posts will be braced to the two nearest line posts with one complete brace assembly to each line post.
- Images are provided for conceptual purpose.

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4.3 Category-4 Fence (Applicable for WAS Admin Area only)

Category - 4 Fencing system shall be provided for this facility as per HCIS 2017 Directive.

Category – 4 Fencing system components are as follows:

4.3.1 Anti-Personnel Fence Area

- The anti-personnel fence shall consist of a 3m high fence configured with one roll of concertina.
- All fence materials including fabric, post and fittings coatings shall conform to HCIS Directive SEC-02.
- Straining posts (corner, end, gate or tension shall be installed at the beginning and end and at a maximum of 90 meters intervals of fence fabric and at changes vertical and horizontal alignments exceeding 22.5° deflection angle. Straining posts shall be braced in the plane of fencing to the adjacent line posts.
- All fence and gate posts shall have minimum yield strength of 42000 psi (290 mpa) and shall meet ASTM A529 requirements as a minimum.

4.3.2 Fence Material Specification

4.3.2.1 Durability Performance

- Fence panels, and associated components, shall be galvanized in accordance with ASTM A641 requirements then fusion bond/polyester coated in accordance to ASTM F668. Galvanizing shall be applied after welding.
- Terminal posts, corner posts, line posts, support braces, brace post, top rail, truss rods to be internally and externally galvanized in accordance to ASTM F1043 and then fusion bond/polyester coated internally and externally in accordance to ASTM F668.
- A minimum coating thickness of 350 micron shall be applied.
- The color of the PVC shall be dark green or black as per ASTM F934 & HCIS SEC-02 Security Fencing Systems.
- Coatings damaged in the field during installation or operations shall be repaired using methods and techniques recommended by the manufacturer.

4.3.2.2 Fence Fabric and Panels

- Anti-personnel fences shall be constructed of Chain-Link (CL) fabric.
- Fabric shall be vinyl-coated chain link with a typical 50 mm diamond mesh. The standard length of roll shall be 15 meters, or greater, and shall be woven continuously without splices.

4.3.2.3 Visibility Index (VI)

- The Visibility Index (VI) may not exceed 45%. This requires that no more than 45% of the fence panel area may be covered with fence wiring and strips.

			
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4.3.2.4 Razor Coil

- Razor Tape (i.e. concertina) materials and configurations shall be in accordance with ASTM F1910 with the following specific provisions:
- Razor tape shall typically be 600 mm diameter (+/- 50 mm). Each loop shall consist of 24(+/- 1) clusters of four needle-sharp barbs on 100mm centers, each barb measuring a minimum of 30mm in length.
- The razor tape shall be secured along its entire length by a minimum 2mm thick rail with spikes on it. Where required, razor tape shall be attached to the fence panel with a minimum 2mm thick bracket at 500mm intervals.
- Where razor tape is placed on the ground, each coil shall be securely anchored to the ground at 1.5 meter intervals.
- The fence posts shall be extended as needed to secure the razor tape.
- Installation of razor tape will follow the practice of ASTM F1911.

4.3.2.5 Barbed Wire

- Six strands of barbed wire shall be installed to secure the razor tape installed on top of the fence.
- Strands shall be spaced uniformly and attached to frame with bands, clips, or eyebolts.
- The strands of barbed wire shall be stretched to remove sag and be anchored firmly to extension arms.

4.3.2.6 Tie Wire

- Tie Wire will have a core of 1.4 mm, minimum, and shall be PVC coated with an outside diameter of 2 mm, minimum.
- Tie wire will comply with the requirements for high security round wire ties specified in ASTM F626.
- Any of the high security tie wire methods specified in ASTM F626 are acceptable.

4.3.2.7 Concrete for Fence

Concrete used for fencing system installation will comply with MD-520-1000-EG-CV-SPC-002 Specification and the following requirements:

- Minimum compressive strength of concrete at 28 days will be 28 Mpa for structural concrete (Maximum water cement ratio will be 0.4) and 14 Mpa for lean concrete.
- Type of cement shall be type-V Sulfate resistant (or) as per geotechnical recommendations.
- Before placing components such as Chain link fabric, rails, tension wire and gates, the concrete will have cured a minimum of 7 days unless the concrete reaches at least 75 percent of its design strength.

			
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- Concrete surface protection coatings are as per geotechnical recommendations.

4.3.2.8 Reinforcement for Fence

- All reinforcement steel bars shall be hot rolled grade 60 (ASTM A615) high tensile deformed billet steel bars.
- Fusion bonded epoxy coated rebars will be used for all structural concrete.

4.3.2.9 Electrical Grounding

Grounding will comply with HCIS SEC-02 Section 6.11 and the following specific requirement:

- Fences which are within 10 meter of an enclosed ground grid or ground loop that is connected to equipment operated at 1000 Volt or greater will not be PVC coated and will be grounded at intervals not exceeding 15 meters to the ground grid or loop. All fences within 3 meters of a ground grid or ground electrode will be bonded at the nearest fence post to the ground grid or ground electrode.
- Fences that pass under a transmission line operating at 69 kV and above will not be PVC coated and be grounded at intervals not exceeding 15 meters on that portion of the fence within 100 meter of the power line.
- Fences that cross over a ground grid or conductors that connect two ground grids will not be PVC coated and will have a bond between the grid or conductors and the nearest post. If the crossing area is extensive, the bond is required every 50 meters.

4.3.2.10 Alignment and Terrain Profiling

- Fence will run in straight lines as far as possible. Where changes of direction are required, the angle will not be less than 90 degrees.
- Fences will match the terrain profile. When required due to terrain the fence panel width may be reduced from the nominal 3 meters width requirement and shorter panel widths be used with fence posts increased in length as needed. This may create a stepped profile on top of the fence. The razor tape will seamlessly cover this profile.
- Where the fence crosses feature such as streams and drainage ditches and where conforming of the fence to the ground contour is impractical, the fence will span the depression.

The space below the bottom of the fence will be closed with extra fence panel or barbed wire. If extra length fence posts are required at such locations, they will be furnished and installed in lieu of standard-length posts as may be required.

- A reasonably smooth profile at the fence line will be provided. The bottom of the fence will not be more than 25 mm (1 inch) above the finished ground line.

			
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4.3.2.11 Post Installation Requirements

- All line posts will be equally spaced at 3 meters intervals or less.
- Posts will typically be 3.6 meter in length or as required by fencing system design.
- Fence line, corner, end, gate, brace or pull posts will be spaced as needed to guarantee fence stability or as required due to changes in vertical or horizontal alignment.
- Posts to be coated both externally and internally.
- The top of the installed post will be covered with a cap that is secured using either a pop rivet, screw or bolt.

4.3.2.12 Chain Link Specific Requirements

The fence will be installed in accordance with ASTM F567 and the specific provisions in this section.

- The fencing fabric will be attached to all line posts by means of tie wire.
- The chain link fence fabric will be installed so that the posts are enclosed.
- The bottom of the chain link fabric will be tied to a continuous concrete anchor.
- Tension bars will be used with fabric bands.
- The bottom edge of fabric will be fastened to the tension wire with wire ties.
- The fabric will be stretched tautly so as not to deflect more than 76mm (3inch) in the center of the fence panel in between the two-line posts. Fabric will be fastened securely to the posts.

4.3.2.13 Top Rail and Tension Wire:

- Fence will be installed with a top rail and bottom tension wire.
- The top rail will be continuous except over drive gates. One expansion/contraction coupling will be used for every 30 meters of rail.
- If it is necessary for a top rail to be omitted, a top tension wire will be installed.
- The tension wire will be secured to the chain link fabric with high-security round tie wire.
- Tension wire will be secured to terminal posts by means of a winding bracket.
- PVC coated galvanized tension wires shall conform to ASTM F1664, Class 2A. Corner and pull posts shall be braced to the two nearest line posts with one complete brace assembly to each line post.
- Images are provided for conceptual purpose.

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HCIS Category 4 Chain Link Fence Structural details

			
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4.4 Anti-Personnel fence (Applicable for WAS Pipeline corridor and conveyor belt corridor)

Fencing system shall be provided for this facility as per HCIS 2017 Directive.

fencing system components are as follows:

4.4.1 Anti-Personnel Fence Area

- The anti-personnel fence shall consist of a 3m high fence configured with one roll of concertina.
- All fence materials including fabric, post and fittings coatings shall conform to HCIS Directive SEC-02.
- Straining posts (corner, end, gate or tension shall be installed at the beginning and end and at a maximum of 90 meters intervals of fence fabric and at changes vertical and horizontal alignments exceeding 22.5° deflection angle. Straining posts shall be braced in the plane of fencing to the adjacent line posts.
- All fence and gate posts shall have minimum yield strength of 42000 psi (290 mpa) and shall meet ASTM A529 requirements as a minimum.

4.4.2 Fence Material Specification

4.4.2.1 Durability Performance

- Fence panels, and associated components, shall be galvanized in accordance with ASTM A641 requirements then fusion bond/polyester coated in accordance to ASTM F668. Galvanizing shall be applied after welding.
- Terminal posts, corner posts, line posts, support braces, brace post, top rail, truss rods to be internally and externally galvanized in accordance to ASTM F1043 and then fusion bond/polyester coated internally and externally in accordance to ASTM F668.
- A minimum coating thickness of 350 micron shall be applied.
- The color of the PVC shall be dark green or black as per ASTM F934 & HCIS SEC-02 Security Fencing Systems.
- Coatings damaged in the field during installation or operations shall be repaired using methods and techniques recommended by the manufacturer.

4.4.2.2 Fence Fabric and Panels

- Anti-personnel fences shall be constructed of Chain-Link (CL) fabric.
- Fabric shall be vinyl-coated chain link with a typical 50 mm diamond mesh. The standard length of roll shall be 15 meters, or greater, and shall be woven continuously without splices.

4.4.2.3 Visibility Index (VI)

- The Visibility Index (VI) may not exceed 45%. This requires that no more than 45% of the fence panel area may be covered with fence wiring and strips.

			
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4.4.2.4 Razor Coil

- Razor Tape (i.e. concertina) materials and configurations shall be in accordance with ASTM F1910 with the following specific provisions:
- Razor tape shall typically be 600 mm diameter (+/- 50 mm). Each loop shall consist of 24(+/- 1) clusters of four needle-sharp barbs on 100mm centers, each barb measuring a minimum of 30mm in length.
- The razor tape shall be secured along its entire length by a minimum 2mm thick rail with spikes on it. Where required, razor tape shall be attached to the fence panel with a minimum 2mm thick bracket at 500mm intervals.
- Where razor tape is placed on the ground, each coil shall be securely anchored to the ground at 1.5 meter intervals.
- The fence posts shall be extended as needed to secure the razor tape.
- Installation of razor tape will follow the practice of ASTM F1911.

4.4.2.5 Barbed Wire

- Six strands of barbed wire shall be installed to secure the razor tape installed on top of the fence.
- Strands shall be spaced uniformly and attached to frame with bands, clips, or eyebolts.
- The strands of barbed wire shall be stretched to remove sag and be anchored firmly to extension arms.

4.4.2.6 Tie Wire

- Tie Wire will have a core of 1.4 mm, minimum, and shall be PVC coated with an outside diameter of 2 mm, minimum.
- Tie wire will comply with the requirements for high security round wire ties specified in ASTM F626.
- Any of the high security tie wire methods specified in ASTM F626 are acceptable.

4.4.2.7 Concrete for Fence

Concrete used for fencing system installation will comply with the following requirements:

- Minimum compressive strength of concrete at 28 days will be 28 Mpa for structural concrete (Maximum water cement ratio will be 0.4) and 14 Mpa for lean concrete.
- Type of cement shall be type-V Sulfate resistant (or) as per geotechnical recommendations.
- Before placing components such as Chain link fabric, rails, tension wire and gates, the concrete will have cured a minimum of 7 days unless the concrete reaches at least 75 percent of its design strength.
- Concrete surface protection coatings are as per geotechnical recommendations.

			
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4.4.2.8 Reinforcement for Fence

- All reinforcement steel bars shall be hot rolled grade 60 (ASTM A615) high tensile deformed billet steel bars.
- Fusion bonded epoxy coated rebars will be used for all structural concrete.

4.4.2.9 Electrical Grounding

Grounding will comply with HCIS SEC-02 Section 6.11 and the following specific requirement:

- Fences which are within 10 meter of an enclosed ground grid or ground loop that is connected to equipment operated at 1000 Volt or greater will not be PVC coated and will be grounded at intervals not exceeding 15 meters to the ground grid or loop. All fences within 3 meters of a ground grid or ground electrode will be bonded at the nearest fence post to the ground grid or ground electrode.
- Fences that pass under a transmission line operating at 69 kV and above will not be PVC coated and be grounded at intervals not exceeding 15 meters on that portion of the fence within 100 meter of the power line.
- Fences that cross over a ground grid or conductors that connect two ground grids will not be PVC coated and will have a bond between the grid or conductors and the nearest post. If the crossing area is extensive, the bond is required every 50 meters.

4.4.2.10 Alignment and Terrain Profiling

- Fence will run in straight lines as far as possible. Where changes of direction are required, the angle will not be less than 90 degrees.
- Fences will match the terrain profile. When required due to terrain the fence panel width may be reduced from the nominal 3 meters width requirement and shorter panel widths be used with fence posts increased in length as needed. This may create a stepped profile on top of the fence. The razor tape will seamlessly cover this profile.
- Where the fence crosses feature such as streams and drainage ditches and where conforming of the fence to the ground contour is impractical, the fence will span the depression.

The space below the bottom of the fence will be closed with extra fence panel or barbed wire. If extra length fence posts are required at such locations, they will be furnished and installed in lieu of standard-length posts as may be required.

- A reasonably smooth profile at the fence line will be provided. The bottom of the fence will not be more than 25 mm (1 inch) above the finished ground line.

			
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4.4.2.11 Post Installation Requirements

- All line posts will be equally spaced at 3 meters intervals or less.
- Posts will typically be 3.6 meter in length or as required by fencing system design.
- Fence line, corner, end, gate, brace or pull posts will be spaced as needed to guarantee fence stability or as required due to changes in vertical or horizontal alignment.
- Posts to be coated both externally and internally.
- The top of the installed post will be covered with a cap that is secured using either a pop rivet, screw or bolt.

4.4.2.12 Chain Link Specific Requirements

The fence will be installed in accordance with ASTM F567 and the specific provisions in this section.

- The fencing fabric will be attached to all line posts by means of tie wire.
- The chain link fence fabric will be installed so that the posts are enclosed.
- The bottom of the chain link fabric will be tied to a continuous concrete anchor.
- Tension bars will be used with fabric bands.
- The bottom edge of fabric will be fastened to the tension wire with wire ties.
- The fabric will be stretched tautly so as not to deflect more than 76mm (3inch) in the center of the fence panel in between the two-line posts. Fabric will be fastened securely to the posts.

4.4.2.13 Top Rail and Tension Wire:

- Fence will be installed with a top rail and bottom tension wire.
- The top rail will be continuous except over drive gates. One expansion/contraction coupling will be used for every 30 meters of rail.
- If it is necessary for a top rail to be omitted, a top tension wire will be installed.
- The tension wire will be secured to the chain link fabric with high security round tie wire.
- Tension wire will be secured to terminal posts by means of a winding bracket.
- PVC coated galvanized tension wires shall conform to ASTM F1664, Class 2A. Corner and pull posts shall be braced to the two nearest line posts with one complete brace assembly to each line post.
- Images are provided for conceptual purpose.

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Anti - Personnel Fence Structural details

			
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4.5 Auxiliary Gate

The auxiliary gate shall provide the capability to shutdown pedestrian or vehicle lanes, or the entire main gate area as needed.

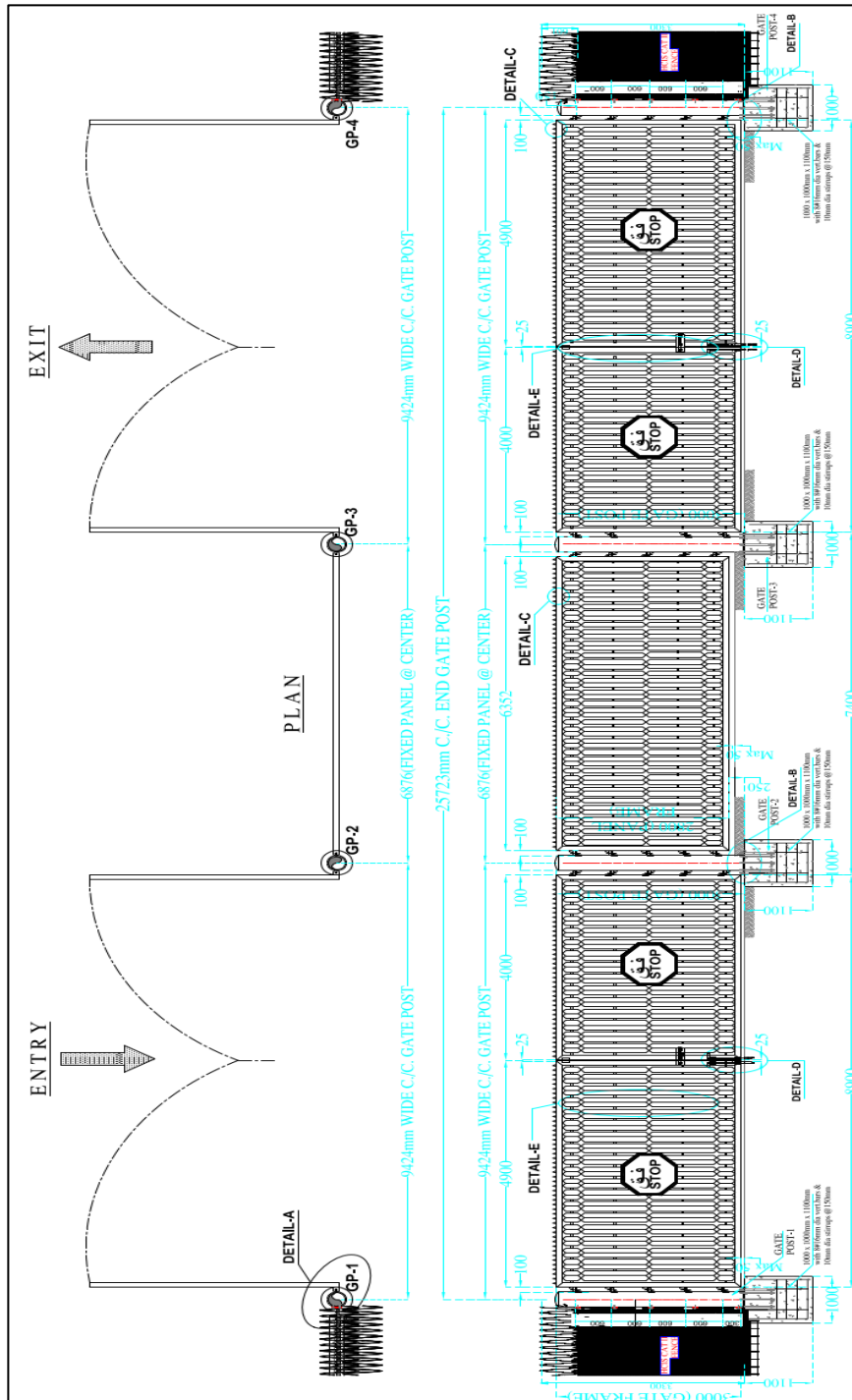
Size of the auxiliary gate shall be based on project requirements.

The auxiliary gate shall meet the specific requirements as per HCIS SEC-03 Directives shown as minimum below:

- When the auxiliary gate is deployed to close a lane or lanes, the AVB's in those lanes must be deployed.
- It shall consist of a steel gate that can be used to close selected entry or exit lanes or to close all lanes when they are not needed.
- Chain link gates shall not be acceptable.
- The auxiliary gate shall be designed to prevent any person entering the facility when the gate is closed.
- The Facility Operator (FO) shall ensure that anti-personnel measures are incorporated in the gate design to prevent an intruder from entering under, through or climbing over the gate when it is closed.
- Gates shall be constructed of strong steel elements that present a reasonable degree of anti-vehicle protection.
- The auxiliary gate shall be at least 3.0m high and shall be able to cover all lanes. Multiple segments shall be used for the gate if needed.
- The auxiliary gate shall be located on the outside of the facility and within 3m - 10m of the gatehouse.
- All gate segments shall have positive locking mechanisms to maintain gate integrity and security.
- Locks used on this gate shall be fully compliant with SEC 09.
- The gate shall use either a motorized sliding or swing mechanism for deployment.
- Clearance between the bottom of the auxiliary gate and the road surface shall at no point be more than 50mm.
- Images are provided for conceptual purpose.

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HCIS - Auxiliary Gate Structural details

			
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4.6 Emergency Gate

emergency Gates are located in the facility perimeter to allow emergency vehicles into the facility or the evacuation of personnel during an emergency incident.

Number of Emergency gates and size shall be based on project requirements

Emergency gates shall meet the specific requirements as per HCIS SEC-03 Directives shown as minimum below:

- Emergency gates shall be designed in accordance with applicable requirements for gates as stipulated in this Directive, section 5.3 & 5.4 of HCIS, Sec-03 as applicable.
- Chain link gates for Class 1 & 2 facilities shall not be acceptable.
- Emergency gates in the facility perimeter of facilities with a FSC of Class 1 or 2 shall be a single steel sliding or swing gate covering the full opening of the gate.
- An emergency gate in the facility perimeter of Class 1 facilities shall be protected with an AVB that meets the requirements of AVB's as stated in SEC-06.
- Concrete jersey barriers or removable bollards shall not be used as anti-vehicle protection for an Emergency Gate. FO shall use crash-rated beams or bollards, as specified in SEC-06, as AVB protection for an emergency gate where applicable.
- FO shall implement a regular maintenance program for all manual or motorized Emergency Gates as specified in SEC-14.
- Emergency gates shall have dedicated video surveillance that is monitored in the local Security Control Centre (SCC).
- Gate width shall be at least 5m to allow the movement of emergency vehicles.
- Emergency gates shall have the same anti-personnel and anti-vehicle characteristics, sensors and surveillance as facility perimeter fencing based on the Facility Security Classification (FSC).
- An Emergency Gate may be designed to accommodate abnormal, large & heavy equipment loads required to occasionally enter the facility.
- Images are provided for conceptual purpose.

